Likelihood ratio tests

Ciaran Evans

Last time

- Optimal (UMP) test of $H_0 : \theta = \theta_0$ vs. $H_A : \theta = \theta_1$ rejects when
$$\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} > k$$
- If this test is the same for all values of $\theta_1 \in \Theta_1$ (depends only on θ_0), then it is UMP for $H_0 : \theta = \theta_0$ vs. $H_A : \theta \in \Theta_1$
- If the distribution has MLR in $T(X)$, then the UMP test of $H_0 : \theta \leq \theta_0$ vs. $H_A : \theta > \theta_0$ rejects when
$$T(X) > t_0$$

In general: Can't find a UMP test for general hypotheses. But we *can* generalize idea of likelihood ratio test (even if it isn't always UMP).

Likelihood ratio test

Suppose we have an iid sample $\mathbf{X} = (X_1, ..., X_n)$ from some distribution with probability function $f(x|\boldsymbol{\theta})$, with parameter $\boldsymbol{\theta} \in \mathbb{R}^d$. We wish to test

$\Theta = \Theta_0 \sqcup \Theta_1$

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \qquad H_A : \boldsymbol{\theta} \in \Theta_1$$

The **likelihood ratio test** (LRT) of these hypotheses rejects H_0 when:

$$\Lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta|X)}{\sup_{\theta \in \Theta} L(\theta|X)} \leq K$$

$\in [0, 1]$

$\leftarrow$ chosen st $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$

Let $\hat{\theta}$ be MLE for $\theta \Rightarrow \sup_{\theta \in \Theta} L(\theta|X) = L(\hat{\theta}|X)$

$$\sup_{\theta \in \Theta_0} L(\theta|X) = L(\hat{\theta}_0|X)$$

$\hat{\theta}_0$ = restricted MLE for θ (in Θ_0, i.e. under H_0)

$$\Lambda = \frac{L(\hat{\theta}_0|X)}{L(\hat{\theta}|X)}$$

Example: comparing linear regression models

Data $(\mathbf{x}_i, Y_i)$ iid from model $Y_i = \mathbf{x}_i^T \beta + \varepsilon_i$, with $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$.

$$\beta = \begin{pmatrix} \beta_{(1)} \\ \beta_{(2)} \end{pmatrix} \qquad x_i = \begin{pmatrix} x_{i(1)} \\ x_{i(2)} \end{pmatrix} \qquad \begin{array}{l} \beta_{(1)}, x_{i(1)} \in \mathbb{R}^{q_1} \\ \beta_{(2)}, x_{i(2)} \in \mathbb{R}^{q_2} \end{array}$$

$$H_0: \beta_{(2)} = 0 \qquad H_A: \beta_{(2)} \neq 0$$

<u>Full model</u> (H_A): $Y_i = x_i^T \beta + \varepsilon_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{q_1-1} x_{i,q_1-1} + \beta_{q_1} x_{i,q_1} + \cdots + \varepsilon_i$

<u>reduced model</u> (H_0): $Y_i = x_{i(1)}^T \beta_{(1)} + \varepsilon_i = \beta_0 + \cdots + \beta_{q_1-1} x_{i,q_1-1} + \varepsilon_i$

$$\theta = (\beta, \sigma^2) \qquad \hat{\theta} = (\hat{\beta}, \hat{\sigma}^2)$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{(unrestricted)}$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - x_i^T \hat{\beta})^2$$

$\frac{1}{n} SSE_{full}$

$$\hat{\theta}_0 = (\hat{\beta}^0, \hat{\sigma}_0^2)$$

$$\hat{\beta}^0 = (X_{(1)}^T X_{(1)})^{-1} X_{(1)}^T Y \quad \text{(restricted)}$$

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - x_{i(1)}^T \hat{\beta}^0)^2$$

$\frac{1}{n} SSE_{reduced}$